

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

WAI4 - KGHS - KOREAN

Issue Date 05-Sep-2019

Revision Date 05-Sep-2019

Revision Number 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Name Iron 1
Product No AC4078-ACT
Kit Reference(s) AC4078
Pure substance/mixture Mixture

Contains Thioglycolic Acid

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent
Uses advised against No Information available

Supplier

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Made in USA

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

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Substances/mixtures corrosive to metal

Category 1

Health hazards

Acute Oral Toxicity

Category 3

Acute Dermal Toxicity

Category 4

Acute Inhalation Toxicity - Vapors

Category 3

Skin Corrosion/Irritation

Category 1

Serious Eye Damage/Eye Irritation

Category 1

Skin Sensitization

Category 1

Environmental hazards

Based on available data, the classification criteria are not met

Label Elements



Signal Word

Danger

Hazard Statements

H290 - May be corrosive to metals

H301 - Toxic if swallowed

H312 - Harmful in contact with skin

H314 - Causes severe skin burns and eye damage

H317 - May cause an allergic skin reaction

H331 - Toxic if inhaled

H318 - Causes serious eye damage

Precautionary Statements

Prevention

P234 - Keep only in original container

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P272 - Contaminated work clothing should not be allowed out of the workplace

P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection

P261 - Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

Response

P303 + P361 + P353 - IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower

P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/ physician

P330 - Rinse mouth

P331 - Do NOT induce vomiting

P363 - Wash contaminated clothing before reuse

P390 - Absorb spillage to prevent material damage

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor/ physician

P321 - Specific treatment (see supplemental first aid instructions on this label)

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P312 - Call a POISON CENTER or doctor/ physician if you feel unwell

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P322 - Specific measures (see supplemental first aid instructions on this label)
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse
P311 - Call a POISON CENTER or doctor/ physician
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting
P333 + P313 - If skin irritation or rash occurs: Get medical advice/ attention
P362 + P364 - Take off all contaminated clothing and wash it before reuse

Storage

P402 - Store in a dry place
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed
P406 - Store in corrosive resistant polypropylene container with a resistant liner
P405 - Store locked up
P406 - Store in corrosive resistant stainless steel container with a resistant liner

Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

Other Hazards

NFPA

Health
3

Flammability
0

Instability
0

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS-No	Index No.	Weight %
Ammonium thioglycolate	No information available	5421-46-5	KE-23090	40 - 50
Thioglycolic Acid	No information available	68-11-1	KE-33786	30 - 40
Water	No information available	7732-18-5	KE-35400	20 - 30

SECTION 4: FIRST AID MEASURES

First aid measures

General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Inhalation

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Move to fresh air. Immediate medical attention is required.

Ingestion

Do not induce vomiting. Call a physician or Poison Control Center immediately.

Self-Protection of the First Aider

Use personal protective equipment. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory

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medical device.

Most important symptoms and effects, both acute and delayed

Most important symptoms and effects Causes burns by all exposure routes May cause allergic skin reaction

Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically

SECTION 5: FIREFIGHTING MEASURES

Suitable Extinguishing Media

CO₂, dry chemical, dry sand, alcohol-resistant foam.

Unsuitable Extinguishing Media

No information available

Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

Explosion Data

Sensitivity to Mechanical Impact None

Sensitivity to Static Discharge None

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Personal Precautions Ensure adequate ventilation. Use personal protective equipment. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

Environmental Precautions Should not be released into the environment. See Section 12 for additional ecological information.

Methods and Material for Containment and Cleaning Up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Handling

Wear personal protective equipment
Do not get in eyes, on skin, or on clothing
Use only under a chemical fume hood
Do not breathe vapors or spray mist
Do not ingest

Conditions for Safe Storage, Including any Incompatibilities

Storage

Corrosives area
Keep containers tightly closed in a dry, cool and well-ventilated place

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Incompatible Products No information available

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS-No	Korea	ACGIH TLV	OSHA PEL
Ammonium thioglycolate	5421-46-5	Not listed	TWA: 1 ppm Skin	Not listed
Thioglycolic Acid	68-11-1	TWA: 1 ppm Skin	TWA: 1 ppm Skin	(Vacated) TWA: 1 ppm (Vacated) TWA: 4 mg/m ³ Skin
Water	7732-18-5	Not listed	Not listed	Not listed

Component	CAS-No	European Union	The United Kingdom	Germany
Ammonium thioglycolate	5421-46-5	Not listed	Not listed	Not listed
Thioglycolic Acid	68-11-1	Not listed	STEL: 3 ppm 15 min STEL: 11.4 mg/m ³ 15 min TWA: 1 ppm 8 hr TWA: 3.8 mg/m ³ 8 hr	Haut
Water	7732-18-5	Not listed	Not listed	Not listed

Biological Exposure Indices

Component	CAS-No	Biological Exposure Indices
Ammonium thioglycolate	5421-46-5	Not listed
Thioglycolic Acid	68-11-1	Not listed
Water	7732-18-5	Not listed

Appropriate engineering controls

Engineering Measures Ensure that eyewash stations and safety showers are close to the workstation location

Individual protection measures, such as personal protective equipment

Personal protective equipment Use only those certified by the Korea Occupational Safety and Health Administration.

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:
Face-shield.

Skin and Body Protection Wear protective gloves/clothing.

Respiratory Protection None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Colorless to pale pink
Odor	Sharp and characteristic
Odor Threshold	No information available
pH	4.5
PH Range	2.5-6.5

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<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling Point/Range	96 °C / 204.8 °F	
Flash Point (High in °C)	125 °C / 257 °F	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble in water	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	
<u>Other Information</u>		
Softening Point	No information available	
Molecular Weight	No information available	
VOC Content(%)	No information available	
Density	No Information available	
Bulk Density	No information available	

SECTION 10: STABILITY AND REACTIVITY

Reactivity

No Information available

Chemical Stability

Stable under normal conditions

Possibility of Hazardous Reactions

None under normal processing

Conditions to Avoid

Extremes of temperature and direct sunlight.

Incompatible Materials

No information available

Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

Product Information

Information on likely routes of exposure

Inhalation	May produce an allergic reaction.
Ingestion	May cause allergic reaction. May be harmful if swallowed.
Eyes	Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. Sensitization.

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Skin

Avoid contact with skin. Causes burns. Skin corrosion/irritation. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. Harmful in contact with skin.

Information on Toxicological Effects

Symptoms

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

(b) skin corrosion/irritation; Sensitization

Category 1. B.
No information available

Mutagenic Effects

No information available

Carcinogenicity

No information available.

Reproductive Effects

No information available

STOT - single exposure

No information available

STOT - repeated exposure Target Organs

No information available
None known.

Aspiration hazard

No information available

Numerical measures of toxicity - Product Information

The following values are calculated based on chapter 3.1 of the GHS document .

ATEmix (oral)	229 mg/kg
ATEmix (dermal)	2375 mg/kg
ATEmix (inhalation-gas)	183 mg/L
ATEmix (inhalation-dust/mist)	1.6 mg/L
ATEmix (inhalation-vapor)	0.7 mg/L

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity

72% of the mixture consists of component(s) of unknown hazards to the aquatic environment

Persistence and Degradability

No information available

Bioaccumulation/ Accumulation

No information available

Mobility

No information available.

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

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Waste Disposal Methods Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport Not Regulated

IATA

UN-No UN1760
Proper Shipping Name Corrosive Liquid, n.o.s. (contains ammonium thioglycolate)
Hazard Class 8
Packing Group II

IMDG/IMO

UN-No UN1760
Proper Shipping Name Corrosive Liquid, n.o.s. (contains ammonium thioglycolate)
Hazard Class 8
Packing Group II
Marine Pollutant No

Special Precautions for User No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS-No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	AICS
Ammonium thioglycolate	5421-46-5	KE-23090	X	226-540-9	X	X	-	X	X	X
Thioglycolic Acid	68-11-1	KE-33786	X	200-677-4	X	X	-	X	X	X
Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	-	X

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Listed	Not applicable	Not applicable
Water	7732-18-5	Listed	Not applicable	Not applicable

Korean National Regulations

Component	CAS-No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Ammonium thioglycolate	5421-46-5	Annex 1 - KE-23090	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Annex 1 - KE-33786	Not applicable	Listed
Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable

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Component	CAS-No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	97-1-68 (>25%)	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable

Component	CAS-No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable

Component	CAS-No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable

Component	CAS-No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	TWA: 1 ppm Skin	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS-No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	Not applicable	Not applicable	5. Group 3 Petroleum (Soluble) 4000 L	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

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Control Parameters

Component	CAS-No	Korea	Biological Exposure Indices
Ammonium thioglycolate	5421-46-5	Not listed	Not listed
Thioglycolic Acid	68-11-1	TWA: 1 ppm Skin	Not listed
Water	7732-18-5	Not listed	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

Component	CAS-No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable

CERCLA Not applicable

Component	CAS-No	CERCLA EHS RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Ammonium thioglycolate	5421-46-5	Not applicable	Not applicable	Not applicable
Thioglycolic Acid	68-11-1	Not applicable	Not applicable	Not applicable
Water	7732-18-5	Not applicable	Not applicable	Not applicable

CLP Classification - Regulation (EC) No 1272/2008

Danger.

H290 - May be corrosive to metals. H317 - May cause an allergic skin reaction. H301 - Toxic if swallowed. H312 - Harmful in contact with skin. H331 - Toxic if inhaled. H314 - Causes severe skin burns and eye damage.

P304 + P340 - IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing. P280 - Wear protective gloves/ protective clothing/ eye protection/ face protection. P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. P303 + P361 + P353 - IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P310 - Immediately call a POISON CENTER or doctor/ physician.

SECTION 16: OTHER INFORMATION

Prepared By Regulatory Affairs

Prepared For Thermo Fisher Scientific Inc.©

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Reason for revision Initial Release.

MOEL's Public Notice No. 2016-19 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

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government requirements. Since conditions of use of the product are not under direct control of Thermo Fisher Scientific, it is the duty of the buyer/user to determine the necessary conditions for the safe use of the product. Thermo Fisher Scientific will not be liable for any injuries or damages resulting from handling, use, misuse or contact with the product.

End of Safety Data Sheet